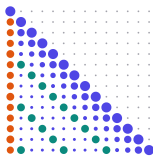


Jiecheng Lu

● NextGen sequence modeling ● Attention meets dynamical systems ● Forecasting for real-world decisions

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Ph.D. candidate in Machine Learning at Georgia Tech advised by Shihao Yang. I develop sequence-model architectures for LLMs and time series that improve **expressivity under practical compute constraints**, spanning attention, linear attention, and dynamic MLPs. My ongoing work studies agentic AI behavior and LLM self-improvement through parameter memory folding. Since 2024, I have published 10 papers as the main contributor at ICLR, ICML, and NeurIPS.

01 EDUCATION

Ph.D. in Machine Learning, Georgia Institute of Technology, Atlanta, GA (Advisor: Shihao Yang) Aug 2023–Apr 2027 (expected)
M.S. in Analytics, Georgia Institute of Technology, Atlanta, GA Aug 2021–Aug 2023
B.S. in Logistics Engineering, Tianjin University, Tianjin, China Sep 2016–Jun 2020

02 RESEARCH INTERESTS

Foundation models and LLM pretraining | Efficient and expressive sequence modeling | Agentic AI | LLM self-improvement
In-context reasoning | Time series analysis | Sequence modeling across NLP, vision, and scientific data

03 SELECTED PUBLICATIONS

* corresponding author

● **NextGen sequence modeling** – LLM pretraining, efficient and expressive attention, and dynamic MLPs

HyperMLP: An Integrated Perspective for Sequence Modeling. ICML 2026
[Jiecheng Lu](#), Shihao Yang.

Free Energy Mixer. ICLR 2026
[Jiecheng Lu](#), Shihao Yang.

ZeroS: Zero-Sum Linear Attention for Efficient Transformers. NeurIPS 2025
[Jiecheng Lu](#), Xu Han, Yan Sun, Viresh Pati, Yubin Kim, Siddhartha Somani, Shihao Yang. + Spotlight

● **Attention meets dynamical systems** – VAR/ARMA attention and symplectic position encoding beyond RoPE

StretchTime: Adaptive Time Series Forecasting via Symplectic Attention. ICML 2026
Yubin Kim, Viresh Pati, Jevon Twitty, Vinh Pham, Shihao Yang, [Jiecheng Lu](#)*

Linear Transformers as VAR Models: Aligning Autoregressive Attention Mechanisms with Autoregressive Forecasting. ICML 2025
[Jiecheng Lu](#), Shihao Yang.

WAVE: Weighted Autoregressive Varying Gate for Time Series Forecasting. ICML 2025
[Jiecheng Lu](#), Xu Han, Yan Sun, Shihao Yang.

● **Forecasting for real-world decisions** – in-context, multivariate, and distribution-valued time series

CAST: Causal Anchored Simplex Transport for Distribution-Valued Time Series. NeurIPS 2026
[Jiecheng Lu](#), Jieqi Di, Runhua Wu, Yuwei Zhou.

In-context Time Series Predictor. ICLR 2025
[Jiecheng Lu](#), Yan Sun, Shihao Yang.

CATS: Enhancing Multivariate Time Series Forecasting by Constructing Auxiliary Time Series as Exogenous Variables. ICML 2024
[Jiecheng Lu](#), Xu Han, Yan Sun, Shihao Yang.

ARM: Refining Multivariate Forecasting with Adaptive Temporal-Contextual Learning. ICLR 2024
[Jiecheng Lu](#), Xu Han, Shihao Yang.

04 RESEARCH TRANSLATION

CDC FluSight forecasting hub Fall 2024–Present
Submit weekly influenza hospitalization forecasts for all U.S. states to the CDC FluSight forecasting hub (entry name: Gatech-ensemble), translating time-series forecasting research into public-health decision support. Co-author of Multi-season evaluation and analysis of categorical trend forecasts of influenza hospital admissions in the United States (medRxiv preprint, 2026; submitted to *Nature Communications*).

05 RESEARCH LEADERSHIP AND MENTORING

Since Fall 2024, I have organized a Georgia Tech undergraduate Sequential AI team, leading each project through the full research cycle, from idea to publication.

<u>Yubin Kim</u>	ZeroS (NeurIPS 2025), StretchTime (ICML 2026); 1 ICLR 2027 submission	→ <u>now Ph.D. student in OR, MIT</u>
<u>Viresh Pati</u>	ZeroS (NeurIPS 2025), StretchTime (ICML 2026); 1 ICLR 2027 submission	→ <u>now Ph.D. student in CS, UPenn</u>
<u>Jevon Twitty</u>	StretchTime (ICML 2026); 3 ICLR 2027 submissions	→ <u>undergraduate, graduating 2027</u>
<u>Vinh Pham</u>	StretchTime (ICML 2026); 3 ICLR 2027 submissions	→ <u>undergraduate, graduating 2028</u>

06 TEACHING AND SERVICE

Independent instructor, *ISyE 4031 Regression and Forecasting*, Georgia Tech Summer 2026

Three sections (one in person, two online), 58 students in 28 project teams. Final Project Showcase; Course Materials. Participant in Georgia Tech's Tech to Teaching program; completed CETL 8717 Course Design for Higher Education.

Reviewing for NeurIPS 2024–2026, ICLR 2025–2026, ICML 2025–2026, and IEEE Internet of Things Journal.

07 SELECTED TALKS

Rethinking Sequence Modeling with HyperMLP: An Integrated Architectural Perspective Mar 2026

Invited Talk, Knowledge Engineering Group, Tsinghua University (hosted by Prof. Jie Tang, founder of Z.ai)

Rethinking Sequence Modeling: LLM Scaling Laws, Expressivity-Efficiency Tradeoffs, and the Role of Architecture Feb & Apr 2026

PhD Seminar, Georgia Tech ISyE PhD Student Seminar (Feb) and Georgia Tech Machine Learning Student Seminar (Apr)

08 HONORS AND AWARDS

George Fellowship, Georgia Tech 2026

Best Poster Award (1st Place), AASF AIX Summit 2026, Poster: “Free Energy Mixer” 2026

Margaret and Stephen Kendrick PhD Student Fellowship for Research Excellence, Georgia Tech 2025

M.H. Stewart Fellowship, Georgia Tech 2023

09 COMPUTING GRANTS

Lambda Research Grant, PI. Mar 2026–Mar 2027

Subquadratic HyperMLP via Convolution-Based Sequence Mixing.

NVIDIA Academic Grant, awarded project. Apr 2026–Sep 2026

Scaling Zero-Sum Linear Attention for Cross-Modal Foundation Models.

NSF ACCESS Program, awarded project. Sep 2025–Sep 2026

Advanced Transformer-based Models for Long-term Time Series Forecasting.

10 PROFESSIONAL EXPERIENCE

Tencent Medical AI Lab (JARVIS Lab) · Data Scientist Intern Jul 2021–Jul 2022

Led development of deep learning and statistical models for medical insurance forecasting and epidemic monitoring; co-authored 2 medical time-series papers, filed 6 patents as first inventor, and delivered technical solutions for medical facilities across 4 major cities handling up to 10 million data points daily.

Amazon Private Brands, Global Sourcing Team · Business Analyst Intern Jan 2021–Jul 2021

Applied machine learning and causal inference to cost analysis and sourcing decisions; built AWS-based dashboard web apps that informed decision-making and reported savings of over USD 100,000 monthly.

Peking University, Guanghua School of Management · Research Assistant Sep 2020–Jan 2021

Country Garden Group · Strategic and Data Analytics Intern Jul 2020–Sep 2020

11 SELECTED PATENTS

Feature Representation Extraction for Time-Series Data. CN115114345B

Data Processing for Attention-Based Time-Series Forecasting. CN114358186B

Training an Anomaly Detection Model for Time-Series Data. CN115130656B

Interpretable Time-Series Forecasting. CN114676176B

Multi-Source Multimodal Data Prediction. CN117010542A

Medical Data Generation from Unordered Inputs. CN117012347A